

Shuo Lv

AI Agents · AI for Science · Cognitive Neuroscience · Complex Networks · Pattern Recognition

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EDUCATION

Beijing Institute of Technology

Master's Student in Electronic Information (085409)

In Progress

Beijing, China

Research interests: AI for Science (cognitive neuroscience), complex networks, and pattern recognition.

Southwest University

Bachelor's Degree in Data Science and Big Data Technology (080910T)

Chongqing, China

Ranked **2/103**; received the **National Scholarship twice**, the Gratitude to Modern Chinese Scientists Scholarship, and the Top-Tier Scholarship.

SELECTED PUBLICATIONS

- **Lv, S.**, Li, J., Yang, R. *et al.* Three parsimonious spatiotemporal patterns in cerebellum reveal individual traits in function and behavior. *Nat Commun* **17**, 6232 (2026). <https://doi.org/10.1038/s41467-026-72931-6>
- **Lv, S.**, Wang, Y., Guo, C. & Zhang, L. Effects of experts on the coupling dynamics of complex contagion of awareness and epidemic spreading. *Nonlinear Dyn* **112**, 2367-2380 (2024). <https://doi.org/10.1007/s11071-023-09146-7>
- Gao, T., Xue, M., Zheng, H., **Lv, S.** *et al.* BrainLMM: A Label-Free Framework for Mapping Multi-Semantic Representation in the Human Visual Cortex. *Proc. AAAI Conf. Artif. Intell.* **40**, 4176-4184 (2026). <https://doi.org/10.1609/aaai.v40i6.42413>
- Guo, C., Wang, Y., **Lv, S.** *et al.* Impact of Sampling Disease Detection and Collective Intervention Measures in Epidemic Transmission. In *Proceedings of the 20th Chinese Conference on Computer Supported Cooperative Work and Social Computing (ChineseCSCW 2025)* (2025). <https://link.springer.com/book/10.1007/978-981-92-0291-1>
- Li, J., Zhang, Y., Wu, X. *et al.* Development of areal-level individualized homologous functional parcellations in youth. *Commun Biol* **8**, 1083 (2025). <https://doi.org/10.1038/s42003-025-08509-7>
- Yang, R., Wu, X., **Lv, S.** *et al.* Normative models of individualized functional brain networks reveal language network expansion in autism. *bioRxiv* (2025). <https://doi.org/10.1101/2025.10.23.684286>

RESEARCH INTERNSHIP

ByteDance, Seed Foundation Model - AI4Math – Research Intern, AI Agent

Nov. 2025 - May 2026

Development

- Built a test-time-scaling multi-agent system on the Seed 2.0 model family for olympiad mathematics. Designed Planner, Solver, and Verifier roles for strategy generation, parallel solving, proof verification, and backtracking; completed **5 iterations** over **100+ problems** and averaged **6.4/7** on an IMO problem set.
- Led an automated Lean 4 proof-QA workflow integrating static rules, code retrieval, and LLM-as-a-judge evaluation. Produced **600+ formalized data items** and improved expert review efficiency by **70%+**.
- Standardized evaluation-data SOPs covering selection, classification, correctness, difficulty, expert annotation, and QA; delivered **2,000+ data items** and authored **11** production and quality specifications.

RESEARCH EXPERIENCE

General-Purpose Cybersecurity Agent – Project Lead <i>Advisor: Libo Zhang, Southwest University</i> • Built 5 specialized agents and a 16-node bounded state machine with LangGraph, separating LLM-based interpretation, planning, and reflection from explicit permissions, routing, and termination rules for code audit, penetration testing, incident response, and reverse engineering. • Designed R0-R3 risk levels, Finding-Evidence validation, human approval for high-risk actions, and a two-layer state/domain blackboard, enabling traceable and evaluable decisions, tool calls, and evidence flows while limiting memory contamination.	<i>Jun. 2026 - Present</i>
Low-Dimensional Spatiotemporal Dynamics of the Human Brain – Project Lead <i>Advisor: Guoyuan Yang, Beijing Institute of Technology</i> • First applied complex principal component analysis to resting-state cerebellar fMRI and identified three dominant sparse patterns jointly capturing zero-lag spatial structure and time-lag dynamics; validated robustness through network features, sequence reconstruction, and modularity analysis. • Applied sCCA and linear SVMs to link individualized cerebellar activity with age, sex, cognition, and emotion. Resulted in 2 manuscripts, including a first-author article in <i>Nature Communications</i> and one manuscript under review.	<i>Dec. 2023 - Present</i>
Complex Social Awareness Modeling – Project Lead <i>Advisor: Libo Zhang, Southwest University</i> • Modeled expert influence and isolation behavior in epidemic prevention using prospect and regret theories. Random-network simulations quantified intervention benefits and revealed phase transitions in social awareness; published 2 papers, including a first-author <i>Nonlinear Dynamics</i> article and a CCF Class C paper.	<i>Apr. 2022 - Nov. 2024</i>
Intelligent Factory Security – Project Lead <i>Advisor: Mingyang Zhong, Southwest University</i> • Developed adaptive multimodal identity recognition for faces under abnormal illumination/occlusion and voices in noise. Used three-way decisions (accept, reject, defer) to exploit visual/audio uncertainty estimates; won a National First Prize in the 2023 China Robot and Artificial Intelligence Competition.	<i>Apr. 2022 - Sep. 2023</i>
Large Language Model for Early Childhood Education – Research Project <i>Advisor: Libo Zhang, Southwest University</i> • Built a personalized multimodal teaching-material generation system on vivo’s BlueLM with content generation, interactive dialogue, and AI-assisted drawing; won a National First Prize in the 2024 China Collegiate Computing Contest - AIGC Innovation Track.	<i>Dec. 2023 - Oct. 2024</i>
Xiangqi Game-Playing System – Project Lead <i>Advisor: Shenglin Li, Southwest University</i> • Implemented a Java Chinese chess engine using a pruned α-β search tree, custom search and evaluation functions, human-vs-computer play, and game-record storage; won a National Second Prize in the 2022 Chinese Computer Game Championship.	<i>Apr. 2021 - Aug. 2021</i>

AWARDS AND HONORS

Jun. 2023	China Robot and Artificial Intelligence Competition	National First Prize
Oct. 2024	China Collegiate Computing Contest - AIGC Innovation Track	National First Prize
Oct. 2023	Contemporary Undergraduate Mathematical Contest in Modeling	National Second Prize
Aug. 2022	Chinese Computer Game Championship	National Second Prize
Sep. 2024	China International College Students’ Innovation Competition	National Third Prize

TECHNICAL SKILLS

Languages	Python, C++, Java, R, Lean 4, JavaScript, SQL, HTML, Bash
Software Stack	FastAPI, Pydantic, LangGraph, Guardrails, React, PyTorch
Tools	Git, Model Context Protocol (MCP), PE, FSL, Connectome Workbench, MATLAB, Adobe Photoshop, Premiere Pro, and InDesign